

武汉大学数学与统计学院 2025-2026 第一学期

抽象代数 期中 ZUOKANG

(时间 120 分钟, 每题 10 分)

一. Let $f = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$. For some integer $1 \leq k < n$, assume the existence of a prime p such that:

(1) $p \mid a_0$ but $p^2 \nmid a_0$

(2) $p \mid a_1, \dots, a_k$

(3) $p \nmid a_{k+1}$

Show that $\exists g \in \mathbb{Z}[x]$ irreducible with $\deg(g) > k$ such that $g \mid f$.

二. Show that

(1) If $I \subseteq R$ is an ideal of R , then $M_n(I)$ is also an ideal of $M_n(R)$.

(2) Every ideal in $M_n(R)$ has the form $M_n(I')$ where I' is an ideal of R .

三. Let $D \equiv 1 \pmod{4}$.

(1) Show that S is a subring of $M_2(\mathbb{Z})$, where

$$S = \left\{ \begin{pmatrix} a & b \\ \frac{D-1}{4}b & a+b \end{pmatrix} : a, b \in \mathbb{Z} \right\}$$

(2) Show that $S \cong \mathbb{Z} \left[\frac{1 + \sqrt{D}}{2} \right]$.

四. Let F be a field and $R_n = F[x]/(x^n)$.

(1) Find all units of R_n .

(2) Find all ideals of R_n .

(3) Show that there exists a unique homomorphism $\pi : R_2 \rightarrow F$ such that $\pi|_F = \text{id}_F$.

- (4) Define $D : F \rightarrow F$ as a derivative: $D(a+b) = D(a) + D(b)$ and $D(ab) = aD(b) + bD(a)$.
Then there exists a bijection:

$$\{D : F \rightarrow F : \text{derivative}\} \simeq \{\phi : F \rightarrow R_2 : \pi \circ \phi = \text{id}_F\}$$

五. Given $\mathbb{Q}/\mathbb{Z} = \{r + \mathbb{Z} : r \in \mathbb{Q}\}$.

- (1) Show that $\left\langle \frac{1}{n} + \mathbb{Z} \right\rangle$ is the unique cyclic group of order n .
- (2) Show that every subgroup generated by finite elements is a cyclic group.
- (3) If H is abelian, show that $\text{Hom}(G, H)$ is a group.
- (4) Show that $\text{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{C}^\times) \cong \text{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$.

六. 暂无信息

七. G is a finite abelian group. Assume that

$$\phi : G \rightarrow G, \quad g \mapsto g^n$$

is a group automorphism.

- (1) $\forall a \in G$, show that

$$\bar{\phi} : G/\langle a \rangle \rightarrow G/\langle a \rangle, \quad \bar{g} \mapsto \bar{g}^n$$

is again a group automorphism.

- (2) Show that $(n, |G|) = 1$.

八. Let $X = \{x, y : x, y \text{ are two 3-cycles in } S_6\}$. Show that

$$\langle X \rangle \cong A_5 \text{ or } C_3 \text{ or } C_3 \times C_3 \text{ or } A_4.$$

九. Let G be a group.

- (1) Let $H, K \leq G$ and $K \trianglelefteq G$, if H and K are solvable, show that HK is solvable
- (2) Let G be a solvable group and $\{e\} \neq H \trianglelefteq G$, show that there exists $K \neq \{e\}, K \leq H$ s.t.
 K is an abelian subgroup of G

†. Let A be a subgroup of G , $N_G(A) := \{g \in G : gAg^{-1} = A\}$ be the normalizer of A in G .

(1) Show that $N_G(A) \leq G$.

(2) Show that $N_G(A)$ is the largest subgroup of G such that $A \trianglelefteq N_G(A)$.

(3) Let $B := \bigcap_{g \in G} g^{-1}Ag$. Show that $B \trianglelefteq A$ and B is the largest subgroup of A such that $B \trianglelefteq G$.

(4) Show that if $[G : A]$ is finite, then $[G : B]$ is also finite.

- **Hint:** Let K be a finite index subgroup of G , show that $\forall H \leq G$, we have $[G : K] \geq [H : H \cap K]$.